Storytelling as Inverse Inverse Planning

Chandra, Li, Tenenbaum & Ragan-Kelley (2023) — with Borges, *The Garden of Forking Paths*

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Master-Seminar: Theory of Mind

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Scan for notes and slides

Motivation

Two triangles and a circle

Heider & Simmel (1944)

A 90-second animation of simple shapes.

- viewers see agents with goals
- they make moral judgements
- no faces, no words, only motion

Can a computer do this?

First try: a rational planner.

- agents reach their goals
- but the motion is flat
- no suspense, surprise or irony

Question of the paper

What does a storyteller do that a rational agent does not?

Animations as visual ToM tests: 19.11. (multimodal ToM benchmark)

Acting *in* vs. acting *out* a situation

Acting *in* a situation

Optimize the character's reward

- agent wants a goal
- best action: shortest path

Acting *out* a situation

Optimize the audience's belief

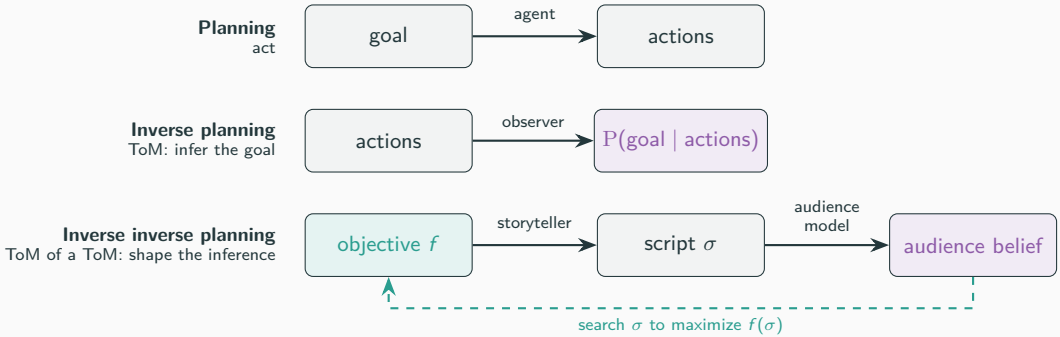
- audience must *infer* the goal
- best action can be sub-optimal

Stories as "probability design" (Kukkonen, 2014)

Good stories design the journey of the reader's beliefs.

Background

Planning, inverse planning, inverse inverse planning



Key idea

The storyteller plans *through* a model of the audience's theory of mind.

Bayesian inverse planning

Theory of mind as Bayes' rule

$$\mathrm{P}(\text{goal} \mid \text{action}) \propto \mathrm{P}(\text{action} \mid \text{goal})\,\mathrm{P}(\text{goal})$$

- likelihood: softmax-rational agent
- prior: plausible goals

Earlier in this seminar

- 03.12.: Bayesian ToM (Baker et al.)
- 17.12.: LLM-augmented inverse planning
- 14.01.: `memo` hands-on

This paper

Inverse planning is not used to *read* minds, but as the audience model.

The audience model: help or hinder?

Grid-world kitchen

- a strong robot, a weak cheese
- cheese moves succeed 60% of the time
- goal tiles: pink, green

Social reward

- robot gets $\rho \cdot r_{\text{cheese}}$ extra
- $\rho > 0$ helps, $\rho < 0$ hinders
- robot models the cheese modelling the robot

Observer

$$\mathrm{P}(H \mid s \to a) \;\propto\; \mathrm{P}(s \to a \mid H)\,\mathrm{P}(H), \qquad H = \langle G_{\text{cheese}}, G_{\text{robot}}, \rho \rangle$$

Audience model from Ullman et al. (2009). Full hypothesis space in the appendix.

Inverse inverse planning

Optimizing a script

🎞 Script

$\sigma = (s_0,\ \sigma_1, \ldots, \sigma_T)$

$\sigma_t = \langle a_t^{\text{robot}}, a_t^{\text{cheese}}, s_t \rangle$

- the setting s_0 is optimized too
- so is the outcome of each move

◎ Objective: depict "helping"

$$f_{\text{help}}(\sigma) = \sum_{t=1}^{T} \mathrm{P}(\rho_{\text{robot}} > 0 \mid \sigma_{1:t})$$

- strong belief at every step
- beam search over scripts

Abstract objective

f_{help} fixes no goals or positions. The optimizer finds a scene that shows helping.

Naïve planning vs. inverse inverse planning

Naïve planning

Figure placeholder:

`assets/fig1_naive.pdf`

Cheese already at pink, robot goes to green. The observer stays unsure.

Inverse inverse planning

Figure placeholder:

`assets/fig2_iip.pdf`

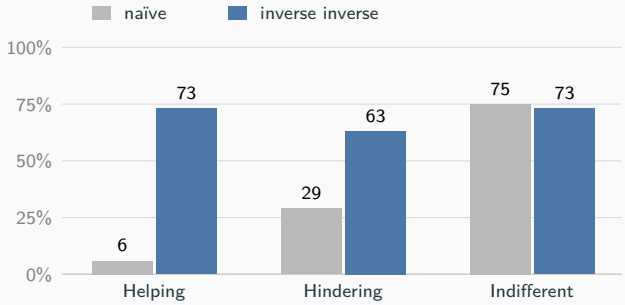
Robot pushes cheese to pink, then steps back. The observer is sure: helping.

based on Chandra et al. (2023), Fig. 1 and Fig. 2.

Behavioral experiment 1: do humans agree?

Setup

- 120 animations: 20 seeds $\times$ 3 relations $\times$ 2 methods
- 98 participants, 15 videos each
- help / hinder / indifferent / unclear



% matching responses. Help and hinder: $p < 0.01$. Indifferent: n.s.

Reading

Indifference is easy for a planner. Helping is hard: the cheese is often close to its goal.

Human vs. model comparison of ToM: 29.10.

Miming by "inverse inverse physics"

A hopper pulls a box uphill

- audience model: intuitive physics, not ToM
- infer box weight from the motion
- make a light box look heavy

Figure placeholder:

`assets/fig4a_hopper.pdf`

based on Chandra et al. (2023), Fig. 4a.

Behavioral experiment 2

- control pairs: chance (50%)
- honest heavy vs. light: 97% correct
- mime vs. honest light: **95.7%** pick the mime
- mime vs. honest *heavy*: **68.6%** pick the mime

Text: 35 participants. Fig. 4b caption: N = 70.

Storytelling devices as objectives (1/2)

Plot twist

$$f_{\text{twist}} = \sum_t \begin{cases} \mathrm{P}(\rho > 0 \mid \sigma_{1:t}) & t \leq T/2 \\ \mathrm{P}(\rho < 0 \mid \sigma_{1:t}) & t > T/2 \end{cases}$$

Robot "helps" the cheese to pink, then traps it.

Dramatic irony

Audience knows what a character does not.

$$f_{\text{irony}} = \sum_t \mathrm{P}(G_c{=}g \mid \sigma_{1:t}) + \mathrm{P}(\rho < 0 \mid \sigma_{1:t}, G_c{=}g) + \mathrm{P}(\rho > 0 \mid \sigma_{1:t}, G_c{=}p)$$

Robot tries to help, but hinders: wrong belief about the goal.

Irony needs a false belief in a character, as in false-belief tasks: 29.10.

Storytelling devices as objectives (2/2)

Flashback

$$f_{\text{flashback}}(\sigma) = f_{\text{help}}\big(c(\sigma)\big)$$

Append a known event to σ. Search for a past that recasts it as helping.

Narrative arc

- robot's fortune: rise, fall, rise (Vonnegut's story shapes)
- KL penalty keeps apparent goals stable
- deus ex machina: a table "falls from the sky"; no character expects it

All examples: same random seed (0). Full f_{arc} in the appendix.

The Garden of Forking Paths

Borges (1941): the plot in one slide

The situation

- Yu Tsun, German spy in England, 1916
- knows the town of a British artillery park
- Captain Madden is hunting him
- the Chief in Berlin only reads newspapers

What he does

- finds “Stephen Albert” in the phone book
- they talk about a labyrinth-novel
- shoots Albert, is arrested
- newspapers report the murder
- the town of Albert is bombed

Reading 1: the murder as inverse inverse planning

Paper	Borges
Storyteller	Yu Tsun
Audience	the Chief in Berlin
Observations	newspaper reports
Target belief	"the target is *Albert*"
Script σ	train, visit, murder, arrest

Acting *out*, not *in*

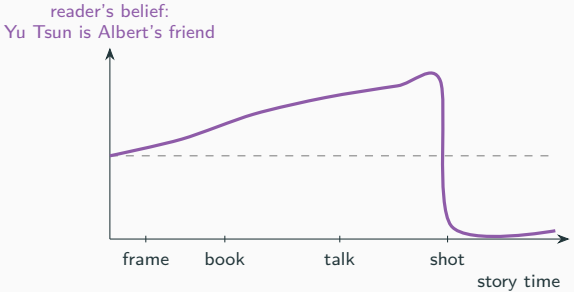
- in the world: kills a friend, gets hanged
- for the belief: the only action that works
- the Chief infers Yu Tsun's goal: ToM

The extreme mime

Like the hopper: a murder that *means* a place name.

Mapping = my reading of the story, not a claim of the paper.

Reading 2: Borges and the reader's beliefs



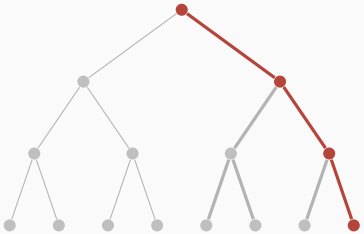
illustrative sketch, not data

🔀 Devices, applied to Borges

- **recasting**: the opening frame gives a known delay new meaning
- **twist**: "friend", then the shot
- **irony**: Albert says in some futures he is the enemy

🔗 Nested beliefs ("the Chief knows that I know ..."): epistemic logic, 04.02.

Reading 3: the garden as the space of scripts



Ts'ui Pen takes *every* branch.

The storyteller keeps one σ.

The chess riddle

- Albert: the novel is about time, but never uses the word
- the reader must infer the concept
- Yu Tsun never sends the word "Albert"

Show, don't tell

Choose one path for its effect on the audience.

Discussion

Limits and critique

From the paper

- scales badly: state, hypothesis, script spaces
- too slow for real-time behavioral experiments
- future work: emotion in the audience model

Raised by Borges

- **shared hypotheses**: the Chief must consider "a killing encodes a name"
- **channel**: one lossy report, not a full trajectory
- **unreliable narrator**: the editor's footnote

Ethics (my extension)

The same machinery can deceive: legible vs. deceptive motion.

Audience aware of being modelled: 07.01. (interlocutor awareness) Recursive ToM: 10.12. invited talk



Questions

Discussion

1. Objective for Yu Tsun: which term makes the murder win?
2. Is the Chief a Bayesian inverse planner? What prior does he need?
3. Can the garden be optimized, or is refusing to choose the point?
4. Speculation: is a good twist a script with a large belief jump (D_{KL})?
5. Could an LLM be the audience model? Clever Hans risk: 05.11.

Sources i


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Action understanding as inverse planning.
Cognition, 113(3):329–349.


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The American Journal of Psychology, 57(2):243–259.


Kukkonen, K. (2014).
Bayesian narrative: Probability, plot and the shape of the fictional world.
Anglia, 132(4):720–739.

Sources ii


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Help or hinder: Bayesian models of social goal inference.
In *Advances in Neural Information Processing Systems*, volume 22.

Appendix: model details

Grid world

- multi-agent MDP, actions $\{\leftarrow, \rightarrow, \uparrow, \downarrow, \text{stay}\}$
- $H = \langle G_{\text{cheese}} \in \{p, g\}, G_{\text{robot}} \in \{p, g, \emptyset\}, \rho \in \{0, \pm 1, \pm 3\}\rangle$
- $\mathrm{P}(s \to a \mid H) \propto \exp\left(\beta\, Q_c^H(s, a)\right)$
- value iteration, beam search, $T = 15$

Physics and arc

- hopper: SHAC, policy $\pi(s; \theta)$, value $V(s; \phi)$
- box weights 0.1 (light), 0.5 (heavy)
- $f_{\text{arc}} = \sum_t \sin\left(\frac{t}{T} 3\pi\right) \mathbb{E}[V_{\text{robot}}^H(s_t) \mid \sigma_{1:t}] - 0.1\, D_{\mathrm{KL}}(H_{1:t-1} \| H_{1:t})$

Related work and code

Planning-based storytelling (Riedl & Young), RSA pragmatics (Frank & Goodman). The audience model could be written in `memo` (14.01.). Code and videos: `https://people.csail.mit.edu/kach/a2i2p/`